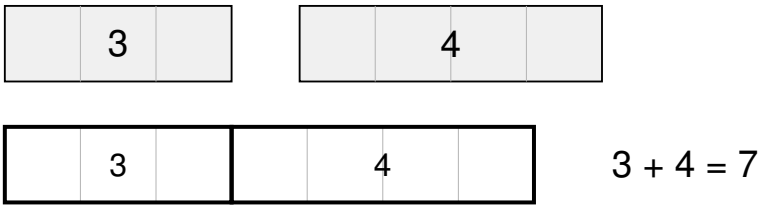
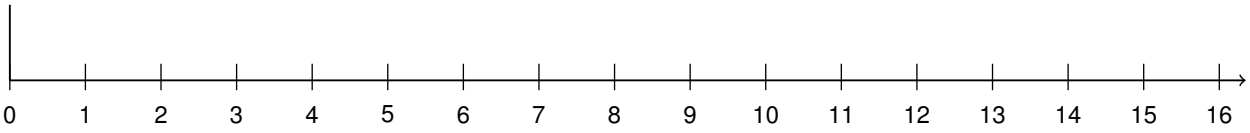
Put whole rods end to end. Leave no gaps or overlaps. You may use a length more than once. Changing the order of the same rods does not make a new way.



Problem 1: Use 3-rods and 4-rods. Find all the lengths from 1 through 16 that can be built, and mark the ones that cannot.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16

Mark a target on a ruler and build from 0 to the mark.



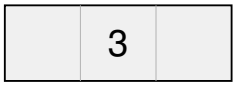
Problem 2: Find every way to make 12, 16 and 24 with 3-rods and 4-rods. Rod order does not count.

12

16

24

Problem 3: Use 3-rods and 5-rods. Find all the impossible lengths from 1 through 20. Is there a last impossible length?



1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	